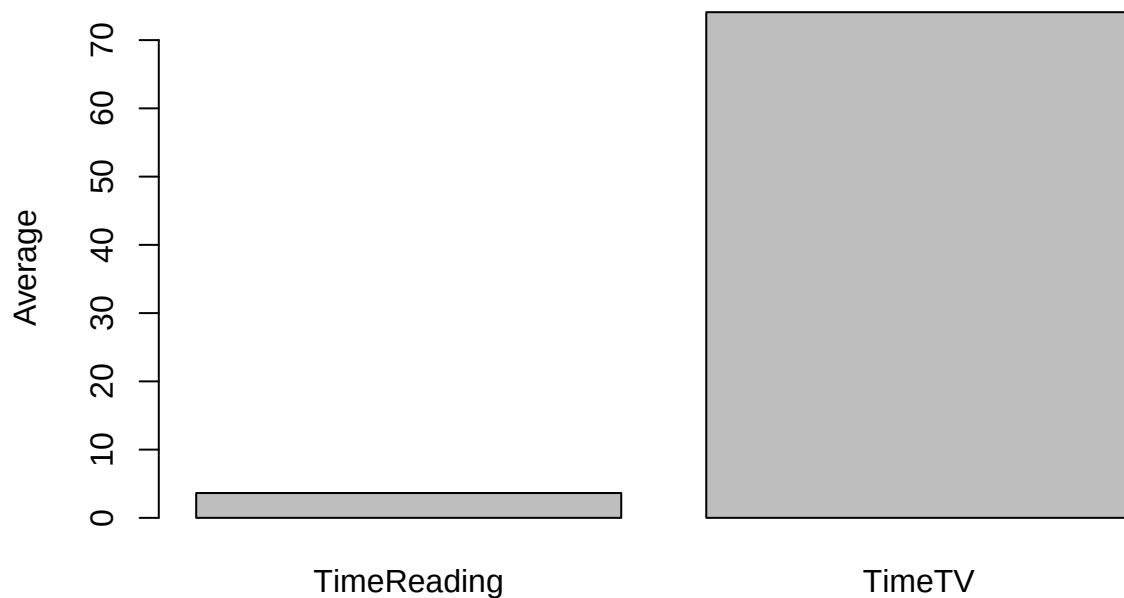
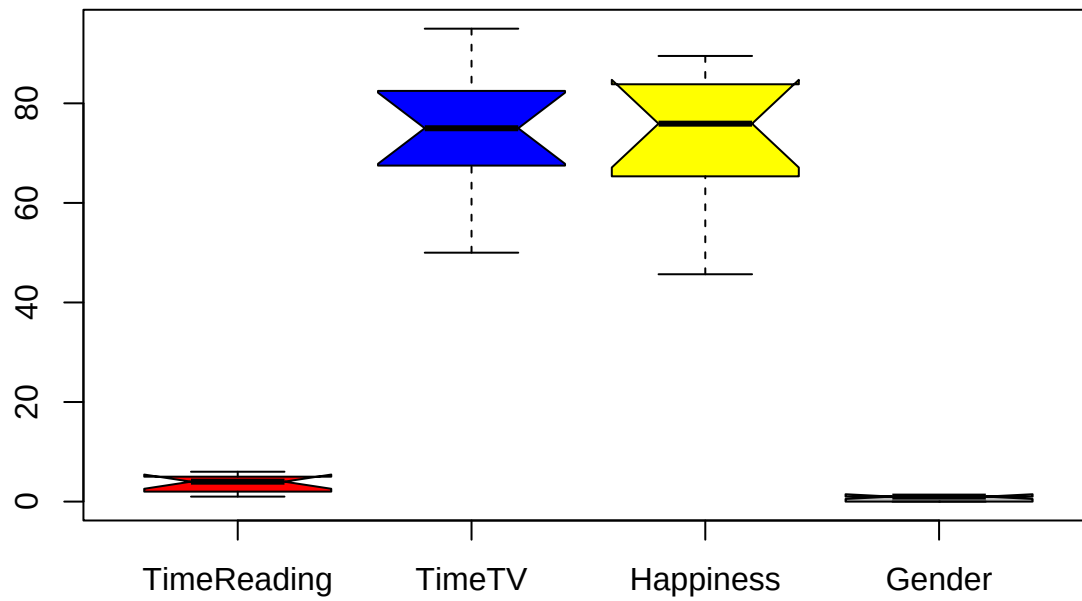


```
##  
## Attaching package: 'dplyr'  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union  
## Read Survey data CSV file for surveys  
df_student_survey = read.csv("student-survey.csv")  
  
# bar chart for average of the 4 quantitative variables  
aveg<- apply(df_student_survey[,1:2], 2, mean)  
barplot(aveg, ylab = "Average")
```

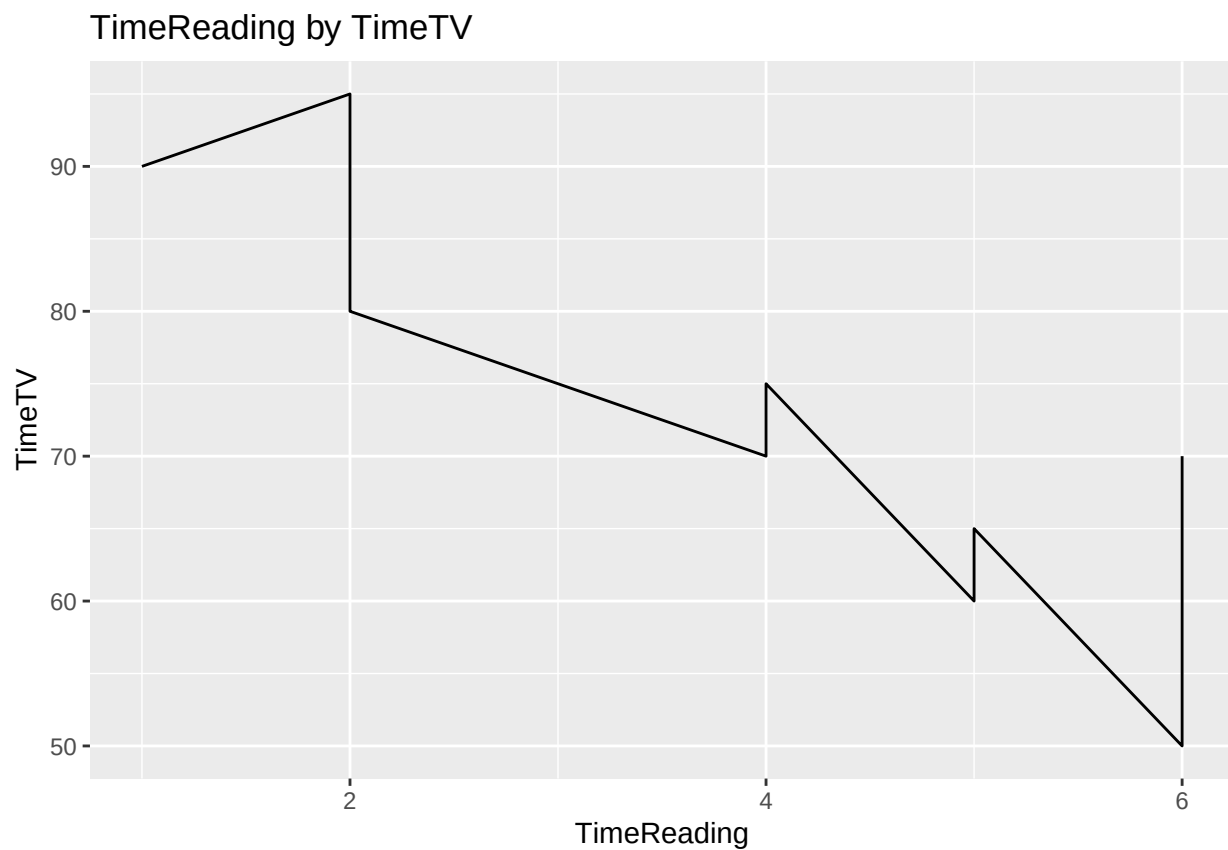


```
## box plot of multiple variables into one figure  
boxplot(df_student_survey[,1:4],notch=T,col=c("red","blue","yellow","grey"))
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, : some  
## notches went outside hinges ('box'): maybe set notch=FALSE
```

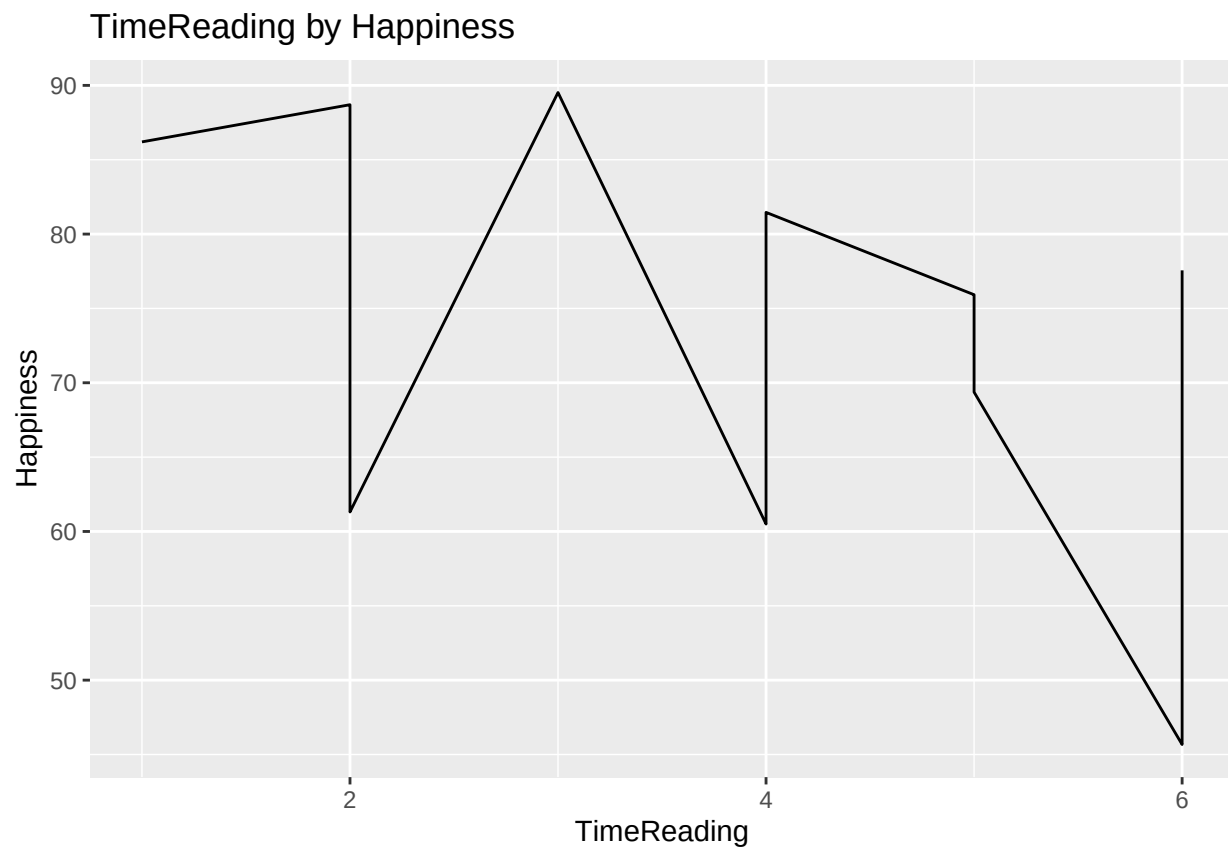


```
# Line Plot TimeReading vs TimeTV
ggplot(df_student_survey, aes(x=TimeReading, y=TimeTV)) +
  geom_line() +
  ggtitle("TimeReading by TimeTV")
```

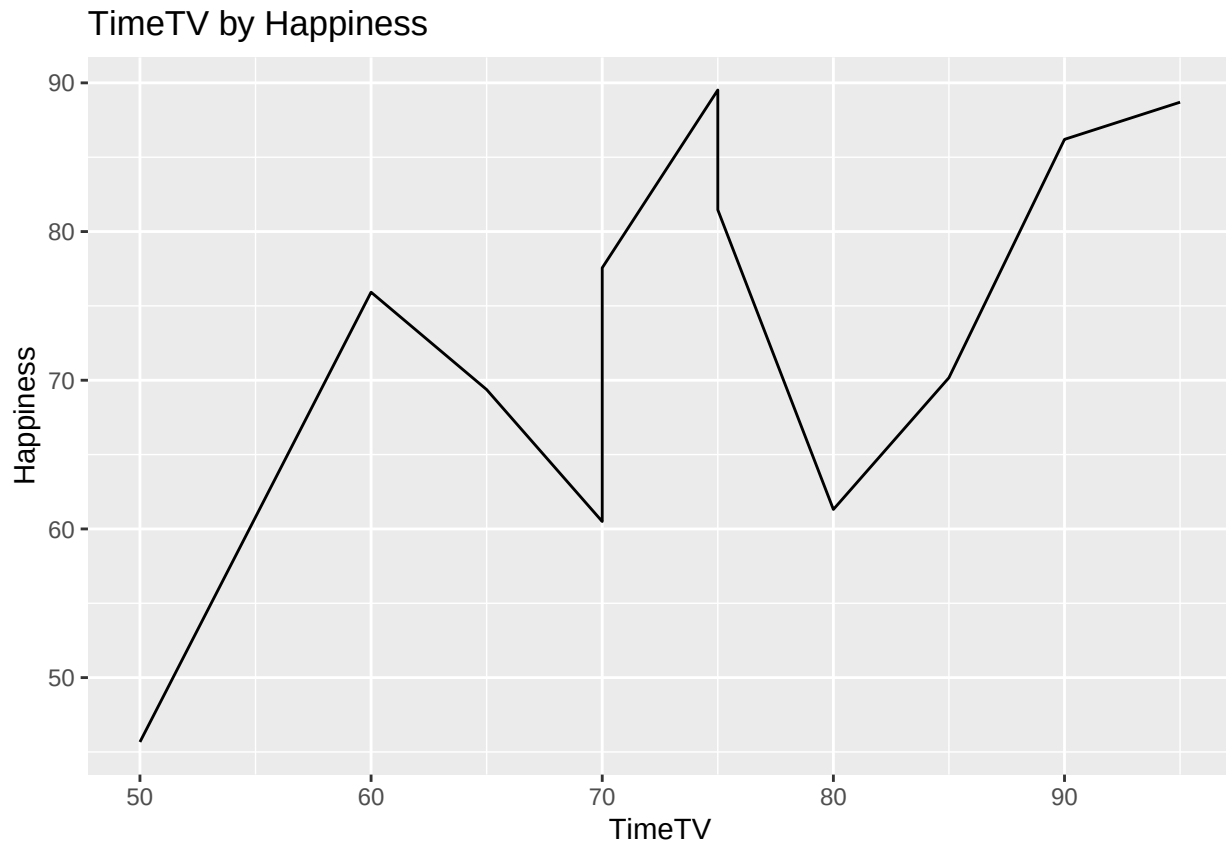


```
# Line Plot TimeReading vs Happiness
ggplot(df_student_survey, aes(x=TimeReading, y=Happiness)) +
  geom_line() +
```

```
ggtitle("TimeReading by Happiness")
```



```
# Line Plot TimeTV vs Happiness  
ggplot(df_student_survey, aes(x=TimeTV, y=Happiness)) +  
  geom_line() +  
  ggtitle("TimeTV by Happiness")
```



```
# covariance matrix with TimeReading, TimeTV and Happiness variables
cov(df_student_survey)
```

```
##           TimeReading      TimeTV  Happiness      Gender
## TimeReading    3.05454545 -20.36363636 -10.350091 -0.08181818
## TimeTV         -20.36363636 174.09090909 114.377273  0.04545455
## Happiness     -10.35009091 114.37727273 185.451422  1.11663636
## Gender         -0.08181818  0.04545455  1.116636  0.27272727
```

4. Survey variables “TimeReading, Happiness” and “TimeTV, Happiness,” each showing the relationship between the time spent reading (TimeReading) or watching TV (TimeTV) on the x-axis and the level of happiness (Happiness) on the y-axis, allowing to visually assess if there is a positive, negative, or no correlation between the time spent on each activity and reported happiness levels.

Positive correlation for reading:

“TimeReading, Happiness” plot shows a positive trend, it shows that people who read more tend to report higher happiness levels.

Negative correlation for TV:

“TimeTV, Happiness” plot shows a negative trend, it indicate that excessive TV watching is associated with lower reported happiness.

```
## correlation matrix for student survey
cor(df_student_survey, use = "complete.obs", method = "pearson")
```

```
##           TimeReading      TimeTV  Happiness      Gender
## TimeReading  1.00000000 -0.883067681 -0.4348663 -0.089642146
## TimeTV      -0.88306768  1.000000000  0.6365560  0.006596673
## Happiness   -0.43486633  0.636555986  1.0000000  0.157011838
## Gender      -0.08964215  0.006596673  0.1570118  1.000000000
```

Between Only Two Variables:

```
## correlation matrix for student survey
cor(df_student_survey$TimeReading, df_student_survey$TimeTV, use = "complete.obs", method = "pearson")

## [1] -0.8830677
```

Between Only Two Variables w/ 99 C.I.:

```
cor.test(df_student_survey$TimeReading, df_student_survey$TimeTV, alternative = "less", method = "pearson")

##
## Pearson's product-moment correlation
##
## data: df_student_survey$TimeReading and df_student_survey$TimeTV
## t = -5.6457, df = 9, p-value = 0.0001577
## alternative hypothesis: true correlation is less than 0
## 99 percent confidence interval:
## -1.0000000 -0.5131843
## sample estimates:
## cor
## -0.8830677
```